Unit 4 - Liquid Rocket Engine

Explanatory notes + quick revision questions

21ASC305T - Rocket Propulsion
Slides ke basis par simple Hinglish me samjhaya hua notes

Unit 4 - Liquid Rocket Engine

Is PDF me repo ke Unit 4 ke dono lectures (L1 aur L2) ka poora content hai: liquid rocket engine ke parts, propellants ke types, propellant ke desirable properties, chemical equilibrium, aur propellant feed systems (gas pressure feed). Sab kuch simple Hinglish me samjhaya hai.

Is PDF me kya kya hai

- Liquid rocket engine ke components aur unke functions (ECU, valves, throttling, actuators)
- Liquid propellants ke types: mono, bi, hypergolic, cold gas, cryogenic, storable, gelled, hybrid
- Boost propulsion vs auxiliary propulsion (table)
- Propellant ke desirable properties (economic, performance, physical, heat transfer)
- Storability aur chemical equilibrium
- Propellant feed systems: gas pressure feed (regulated aur blow-down), pressurizing gas ke 6 sources
- Last me quick revision questions

1. Liquid propellant rocket engine - parts aur kaam

Liquid rocket engine me ye saare parts hote hain:

- **Thrust chamber(s):** jahan propellants jalte hain aur thrust banta hai.
- **Feed mechanism:** jo propellants ko tanks se thrust chamber tak pahunchata hai.
- **Power source:** feed mechanism ko chalane ke liye energy deta hai.
- **Plumbing ya piping:** pipes jinse propellant pressure me transfer hota hai.
- **Structure:** jo thrust force ko vehicle tak transmit karta hai.
- **Control devices (valves):** engine ko start aur stop karne ke liye, aur kabhi kabhi propellant flow (aur isliye thrust) vary karne ke liye.

Control system - ECU aur valves

Propellant control system, valves aur actuation systems ko milakar aksar **thrust control system** ya **Engine Control Unit (ECU)** kehte hain. ECU (ya flight computer) lagataar calculate karta rehta hai ki rocket ki current speed, altitude, weight aur trajectory ke hisaab se kitna thrust chahiye.

Part	Kaam
Propellant control valves (main valves)	Ye physically khulte ya band hote hain taaki fuel aur oxidizer ka flow start ya stop ho.
Throttling valves / cavitating venturis	Flow rate vary karne ke liye, jisse thrust badalta hai. Isse hi **thrust throttling** kehte hain. Isme variable-geometry valves ya adjustable flow controllers use hote hain.
Actuators	Mechanical, hydraulic ya electrical drivers jo flight computer ke command ki exact position par valves ko hilate hain.
Pressure sensors aur flow meters	Engine ke andar actual propellant flow monitor karte rehte hain. Thrust kam ya zyada ho to computer milliseconds me error pakad leta hai.

2. Liquid propellants ke types

Liquid propellant ka matlab wo sab propellants jo liquid form me store hote hain. Ye in me se koi ek ho sakta hai:

- **Oxidizer:** liquid oxygen, nitric acid, nitrogen tetroxide etc.
- **Fuel:** kerosene, alcohol, liquid hydrogen etc.
- **Chemical compound (ya oxidizer-fuel ingredients ka mixture)** jo khud decompose ho sake, jaise hydrazine. Ye bina alag fuel ya oxidizer ke khud toot kar energy release kar deta hai.

- Upar me se koi bhi, par **gelling agent** ke saath (ye abhi production ke liye approved nahi hain). Jaise gelled kerosene, gelled LOX, gelled hydrazine.

Boost propulsion vs Auxiliary propulsion

Slide ki table se (Typical characteristics of two categories of liquid propellant rocket engines):

Feature	Boost propulsion	Auxiliary propulsion
Mission	Vehicle ko flight path par significant velocity dena	Attitude control, minor space maneuvers, trajectory correction, orbit maintenance
Applications	Launch vehicles ke booster aur upper stages, bade missiles	Spacecraft, satellites, antiballistic missile ka top stage, space rendezvous
Total impulse	High	Low
Thrust chambers per engine	Usually 1; kabhi 4, 3 ya 2	4 se 24
Thrust per chamber	High: 4500 N se 7,900,000 N	Small: 0.001 se 4500 N
Feed system	Mostly turbopump; chhote thrust ke liye kabhi pressurized feed	Pressurized feed system with high-pressure gas supply
Tank pressure	0.138 - 0.379 MPa (20-55 psi)	0.689 - 17.23 MPa (100-2500 psi)
Common cooling	Propellant cooled	Radiation cooled
Propellants	Cryogenic aur storable liquids	Storable liquids, monopropellants, stored cold gas
Chamber pressure	2.4 - 21 MPa (350-3600 psi)	0.14 - 2.1 MPa (20-400 psi)
Starts in one mission	Usually no restart; kabhi 1, kabhi 4 tak	Kuch space missions me kai hazaar starts
Total firing duration	Kuch minutes tak	Kuch ghante tak
Shortest firing duration	Typically 5-40 s	Typically 0.02 s (pulsing small thrusters)
Time to reach full thrust	Kuch seconds tak	Bahut fast, 0.004 - 0.080 s
Life in space	Hours, days ya months	15 saal ya usse zyada

Yaad rakho

Table ka short logic: **boost** = bada thrust, high pressure, turbopump, kam restarts. **Auxiliary** = chhota thrust, pressurized gas feed, kai hazaar restarts, bahut tez response.

Monopropellant

- Ek hi liquid me oxidizing agent aur combustible matter dono hote hain.
- Ye kai compounds ka stored mixture ho sakta hai, ya homogeneous material jaise hydrogen peroxide ya hydrazine.
- Normal storage conditions me stable hota hai, par chamber me heat ya catalyst milne par decompose hokar hot combustion gas deta hai.
- Monopropellant gas pressure feed systems simple hote hain kyunki propellant ek hi hai, to pipes, valves aur tanks kam lagte hain.

Bipropellant

- Do alag liquid propellants: ek oxidizer aur ek fuel. Dono alag store hote hain aur combustion chamber ke andar mix hote hain.
- **Hypergolic bipropellant:** oxidizer aur fuel ke contact me aate hi apne aap ignite ho jaate hain. Alag ignition system nahi chahiye.

- **Non-hypergolic bipropellant:** jalane ke liye energy chahiye (jaise electric discharge se heat), isliye ignition system zaroori hai.

Cold gas propellant

- Gas (jaise helium, argon ya gaseous nitrogen) normal temperature par high pressure me store hoti hai.
- Performance comparatively low hota hai, par system simple aur bahut reliable hota hai.
- Roll control aur attitude control ke liye use hui hai.

Cold gas ke uses (slide se): satellite attitude control, station keeping (satellite ko uski orbital slot me rakhna), CubeSats aur nano-satellites (jahan weight aur volume ka budget tight hota hai), spacecraft reaction control systems (docking ke time chhote movements), manned maneuvering units (HHMU, MMU jo NASA astronauts spacewalk me use karte the), aur launch vehicle pressurization support (high-pressure helium gas jo liquid propellants ko combustion chamber me dhakelti hai).

Cryogenic propellant

- Liquefied gas jo ambient se bahut kam temperature par hoti hai. Jaise liquid oxygen (-183 C) aur liquid hydrogen (-253 C).
- Storage tank ko vent karne ki provision aur vaporization losses kam karne ke tareeke zaroori hain.

Storable aur space-storable propellants

- **Storable** (jaise nitric acid ya gasoline): ambient temperature aur modest pressure par liquid rehte hain aur sealed tanks me bahut samay tak store ho sakte hain.
- **Space-storable:** space environment me liquid rehte hain. Inki storability tank design, thermal conditions aur tank pressure par depend karti hai. Example: ammonia.

Gelled propellant

- Gelling additive wala **thixotropic** liquid. Rest me jelly ya gaadhe paint jaisa behave karta hai (aasani se spill ya leak nahi hota), par pressure lagane par behta hai aur jalta bhi hai. Isliye kuch cases me safer hai.
- Gelling agents ko rheological modifiers bhi kehte hain. Gelled fuels ke examples: organometallic compounds, polymeric thickeners, silica-based materials, kuch metal soaps aur surfactant-type additives.
- Jaise hydrocarbon fuel ko polymeric ya particulate gelling agent se gaadha kar sakte hain. Rest me thick paint jaisa, par pipes, valves, pumps ya injectors se pressure ya shear stress lagne par flow karta hai.
- Kuch experimental engines me use hue hain, par abhi tak production me nahi hain.

Hybrid propellant

- Usually liquid oxidizer aur solid fuel.

Type	Ek line me
Monopropellant	Ek hi liquid me fuel aur oxidizer, heat ya catalyst se decompose
Bipropellant	Alag oxidizer aur fuel, chamber me mix
Hypergolic	Contact par apne aap jal jaata hai
Cold gas	High pressure me stored gas, low performance par reliable
Cryogenic	Bahut thanda liquefied gas (LOX -183 C, LH2 -253 C)
Storable	Normal temperature par liquid, lambe time ke liye store hota hai
Space-storable	Space me liquid rehta hai (e.g. ammonia)
Gelled	Thixotropic jelly jaisa, safe, abhi production me nahi
Hybrid	Liquid oxidizer + solid fuel

3. Liquid propellants ki properties

Pehle ye samajh lo ki **liquid propellants ki properties** (reactants, yaani fuel aur oxidizer jaisa wo hain) alag hoti hain, aur chamber me react hone ke baad bani **hot gas mixture ki properties** alag. Liquid propellant ka chemical nature aur uska mixture ratio dono ki properties decide karte hain. Koi bhi practical propellant sabhi desirable properties ek saath nahi deta, isliye propellant selection hamesha ek compromise hota hai.

(a) Economic factors

- Zyada matra me available hona aur low cost bahut zaroori hain.
- Military applications me production, supply aur storage ki logistics bhi dekhni padti hai.
- Production process ke liye ordinary chemical equipment aur available raw materials hi kaafi hone chahiye.
- Toxic ya cryogenic propellants storable aur non-toxic se mehnge padte hain, kyunki inme extra operation steps, extra safety provisions, extra design features, launch se pehle lambe check-out procedures aur aksar better trained personnel chahiye.

(b) Performance of propellants

- Rocket engines ki performance specific impulse, exhaust velocity, characteristic velocity ya doosre engine parameters se compare ki jaati hai.
- Specific impulse aur exhaust velocity in par depend karte hain: **pressure ratio, specific heat ratio, combustion temperature, mixture ratio aur molecular mass**.
- Flight performance parameters me bhi express karte hain. Average propellant density, total impulse aur engine mass ratio flight relations me aate hain.
- High performance ke liye propellant mixture me chemical energy zyada honi chahiye (isse high chamber temperature milta hai).
- Combustion products ki **molecular mass kam** honi chahiye.
- Ye hydrogen-rich fuels se milta hai, jab hydrogen gas ka kaafi hissa uncombined rehta hai.
- Isliye kai bipropellants ke liye best mixture ratio **stoichiometric nahi** hota (stoichiometric me complete oxidation hoti hai aur highest flame temperature milta hai), balki **fuel-rich** hota hai jisme bahut saare low-molecular-mass products bante hain.

> **Yaad rakho**
>
> Exam point: best mixture ratio = **fuel-rich** (stoichiometric nahi), kyunki isme molecular mass kam hoti hai aur Isp badhta hai, bhale hi flame temperature thoda kam ho.

(c) Desirable physical properties

Property	Kyun chahiye
Low freezing point	Taaki rocket thande environment me bhi chal sake. Thode freezing-point depressants daalne se kai liquids ka freezing point kam ho jaata hai.
High specific gravity (density)	Denser propellant se same tank volume me zyada mass aati hai. Ya same mass ke liye chhota tank, kam structural mass aur kam aerodynamic drag. Isse atmosphere me udne wale vehicle ki max velocity aur range badhti hai.
Stability	Storage ya operation me decomposition na ho, high temperature par bhi. Kai propellants 15 saal se zyada store hue hain bina significant deterioration ke. Tubing, pipes, tank walls, valve seats aur gaskets ke contact me chemical deterioration nahi hona chahiye. Moisture absorb nahi karna chahiye aur thodi impurities ka bura asar nahi hona chahiye.
Heat transfer properties	High specific heat, high thermal conductivity, low freezing temperature aur high boiling ya decomposition temperature, ye sab thrust chamber ko cool karne wale propellant ke liye desirable hain.

Stability ke saath ek khaas point: regeneratively cooled thrust chamber ke garam cooling-jacket passages me flow karte time propellant ka koi chemical deterioration nahi hona chahiye. Agar carbon-containing coolant decompose ho kar passages ki garam inside surface par carbonaceous deposits bana de to ye deposits **harden** ho jaate hain, heat flow kam kar dete hain, local metal temperature badha dete hain, aur chamber kamzor hokar fail ho sakta hai.

Heat absorb karne ki formula (slide me):

$$Q = m\, c_p\, \Delta T$$

Jahan,

Q	=	Absorb ki gayi heat, J
m	=	Coolant propellant ka mass, kg
c_p	=	Specific heat, J/kg.K
ΔT	=	Temperature rise, K

Matlab specific heat zyada ho to same temperature rise ke liye propellant zyada heat absorb karta hai, isliye wo acha coolant hai.

4. Storability

Kuch pressure feed systems ko factory me hi storable propellant aur pressurizing agent se pehle se bhar ke rakha ja sakta hai, aur zaroorat par turant operate kiya ja sakta hai. Solid propellant rocket units ke muqabale ye prepackaged liquid propellant pressurized feed systems me ye advantages hote hain:

- Long-term storability
- Random restarts (kabhi bhi dobara start kar sakte hain)
- Transportation vibration ya shock ke against behtar resistance

5. Chemical equilibrium

Chemical equilibrium reversible chemical reactions me tab aata hai jab products banne ki rate bilkul wahi ho jaati hai jo reverse reaction (products se wapas reactants banne) ki rate hai. Equilibrium aane ke baad concentrations me aur change nahi hota.

- Chamber me present gases ki relative proportions pressure, temperature aur mixture ke equilibrium state par depend karti hain.
- Combustion process normally combustion chamber me lagbhag **constant pressure (isobaric)** par hota hai aur resulting gases **Dalton's law** follow karti hain.
- Propellant combustion ke dauran chemical reactions bahut tez hoti hain.
- Chamber volume itna bada maana jaata hai aur residence time itna lamba ki chamber ke andar chemical equilibrium ban jaaye.

6. Propellant feed systems

Feed system ke 2 principal functions hain:

- Propellants ka pressure badhana.
- Unhe design mass flow rate par ek ya zyada thrust chambers tak supply karna.

In kaamo ki energy ya to **high-pressure gas** se aati hai, ya **centrifugal pumps** se, ya dono ke combination se.

Feed system ka selection kis par depend karta hai

- Rocket application aur mission

- Duration
- Thrust chambers ki number ya type
- Past experience
- Design ki simplicity, manufacturing ki aasani, low cost aur minimum inert mass

Gas pressure feed vs turbopump feed

Gas pressure feed systems turbopump systems se **behtar** performance dete hain jab:

- Vehicle ka total impulse ya propellant mass relatively low ho.
- Chamber pressure relatively low ho.
- Engine thrust-to-weight ratio low ho (usually 0.6 se kam).

Gas pressure system me heavy-walled propellant tanks aur pressurizing gas hi engine system ka major inert mass hote hain.

Turbopump feed system me propellant tank pressure **10 se 40 guna kam** hota hai, aur isliye vehicle ke tank masses bhi usi factor se kam ho jaate hain.

	Gas pressure feed	Turbopump feed
Kab behtar	Low total impulse, low chamber pressure, low thrust-to-weight (< 0.6)	Bade impulse aur high chamber pressure ke liye
Tank pressure	High (tank ko heavy-walled banana padta hai)	10 se 40 guna kam
Tank mass	Zyada	Kaafi kam (usi factor se)
Energy source	High-pressure gas	Centrifugal pumps (turbine driven)

7. Gas pressure feed systems

Liquid propellants ko tanks se bahar dhakelne ka sabse simple aur common tareeka high-pressure gas hai. Aaj bhi do common types use hote hain:

Type 1: Regulated gas pressure feed system

- Gas feed line me ek **gas pressure regulator** hota hai. Isse engine lagbhag constant tank pressure par aur lagbhag constant thrust par chalta hai.
- Main components: high-pressure gas tank, gas starting valve, gas pressure regulator, propellant tanks, propellant valves aur feed lines.

Extra components bhi ho sakte hain: filling aur draining provisions, check valves, filters, flexible elastic bladders (liquid ko pressurizing gas se alag rakhne ke liye), pressure sensors ya gauges.

- Jab propellants poore consume ho jaate hain to pressurizing gas ka use lines aur valves se bacha hua liquid propellant residue saaf karne (scavenge) ke liye bhi kar sakte hain.
- Agar unit ko baar baar use ya fly karna ho (jaise space-maneuver rocket) to usme kuch extra features ho sakte hain, jaise thrust-regulating device aur tank level gauge.

Type 2: Blow-down feed system

- Propellant tanks **bade** hote hain kyunki unme propellants ke saath pressurizing gas bhi initial maximum tank pressure par store hoti hai.
- Alag high-pressure gas tank aur pressure regulator **nahi** hote.
- Tank ke andar pehle se bhari gas ka expansion hi propellants ko bahar dhakelta hai. (Isliye tank pressure aur thrust dheere dheere girte jaate hain.)

	Regulated system	Blow-down system
Gas storage	Alag high-pressure gas tank	Propellant tank ke andar hi gas
Regulator	Hota hai	Nahi hota
Tank size	Propellant tanks chhote	Propellant tanks bade

	Regulated system	Blow-down system
Tank pressure / thrust	Lagbhag constant	Burn ke saath girta hai
Complexity	Zyada parts	Simple

Pressurizing gas ke 6 sources

- **High-pressure inert gases** jo ambient temperature par stored hon. Sabse common. Typical gases: helium, nitrogen aur air.
- **Heated high-pressure inert gases** (usually 200 se 800 F ya 93 se 427 C). Isse required gas aur isliye pressurizing system ka inert mass kam hota hai.
- **Chemical reaction se bani gas** (liquid bipropellants ya monopropellant ya solid propellant se), aise mixture ratio par jisse 'warm gas' bane. Chemically generated warm gas wale tank pressurization systems usually heated inert gas systems se halke hote hain, khaas kar high total impulse applications me.
- **Cryogenic liquid propellant ka evaporation:** usually liquid hydrogen ya liquid oxygen ka chhota sa hissa evaporate karte hain, heat thrust chamber ke cooling jacket se ya turbine exhaust gases se (heat exchangers ke saath) le kar. Phir is evaporated flow ka kuch ya poora hissa tank pressurization ke liye use hota hai.
- **Direct injection:** chhoti hypergolic fuel stream ko main oxidizer tank me aur chhoti hypergolic oxidizer stream ko fuel tank me inject karna. Kai countries ne try kiya hai, par limited success mili.
- **Self-pressurization:** cryogenic propellants ko evaporation se khud pressurize karna. Feasible hai par control karna mushkil ho sakta hai.

Pressurizing system design karne ke liye jaruri information

Koi bhi pressurizing system design ya analyze karne ke liye tank aur engine ki ye information chahiye hoti hai (inme se kai ki nominal, maximum aur minimum values chahiye):

- Propellant flow, thrust, duration, pulse width
- Propellant tank volume aur tank volume ka percent ullage (tank me khali jagah jo gas ke liye rehti hai)
- Storage temperature range
- Propellant aur pressurizing gas ki properties
- Propellant tank pressure aur gas tank pressure
- Unavailable residual propellant ki matra (jo tank me reh jaata hai aur use nahi ho paata)

Quick Revision Questions - Unit 4

Unit 4 ke slides me koi numerical question nahi diya gaya hai, isliye numericals ki jagah neeche slides ke content se bane important theory questions ke short answers diye hain. Ye 2 mark / 5 mark ke type ke hain.

Q1. Liquid rocket engine ke main parts likho.

Ans: Thrust chamber, propellants ko tank se chamber tak pahunchane wala feed mechanism, feed mechanism ki power source, plumbing, thrust transmit karne wala structure, aur start/stop/throttle karne wale control devices (valves). Control system ECU ke under aata hai.

Q2. Monopropellant aur bipropellant me kya difference hai?

Ans: Monopropellant me ek hi liquid hota hai jisme fuel aur oxidizer dono hote hain; heat ya catalyst se decompose hota hai (e.g. hydrazine, hydrogen peroxide). Bipropellant me oxidizer aur fuel alag store hote hain aur chamber me mix hote hain.

Q3. Hypergolic aur non-hypergolic propellant kya hai?

Ans: Hypergolic combination contact hote hi apne aap ignite ho jaata hai. Non-hypergolic ko start karne ke liye energy (jaise electric discharge) aur ignition system chahiye.

Q4. Cold gas propellant kya hai aur kahan use hota hai?

Ans: Helium, argon ya nitrogen jaisi gas ambient temperature par high pressure me stored. Low performance par simple aur reliable. Attitude control, roll control, station keeping, CubeSats, RCS aur propellant pressurization me use hota hai.

Q5. Cryogenic propellant kya hai?

Ans: Liquefied gas jo ambient se bahut kam temperature par hoti hai, jaise LOX (-183 C) aur LH2 (-253 C). Tank venting aur vaporization losses kam karne ki provision chahiye.

Q6. Gelled propellant kya hai?

Ans: Gelling agent wala thixotropic liquid. Rest me jelly jaisa (spill nahi hota), pressure par flow karta hai aur jalta hai. Safer hai, par abhi production me nahi.

Q7. Liquid propellant ki desirable properties batao.

Ans: Low freezing point, high specific gravity, chemical stability, accha heat transfer (high specific heat, high thermal conductivity, high boiling point), low cost aur easy availability.

Q8. Best mixture ratio stoichiometric kyun nahi hota?

Ans: Stoichiometric par flame temperature sabse zyada hota hai, par fuel-rich mixture me low molecular mass wale products (hydrogen-rich) bante hain jisse Isp badhta hai.

Q9. Gas pressure feed system kab turbopump se behtar hai?

Ans: Jab total impulse ya propellant mass kam ho, chamber pressure low ho aur engine thrust-to-weight ratio 0.6 se kam ho.

Q10. Regulated aur blow-down gas pressure feed me difference batao.

Ans: Regulated system me alag high-pressure gas tank aur pressure regulator hota hai, tank pressure aur thrust lagbhag constant. Blow-down me tanks bade hote hain, gas propellant ke saath hi tank me stored hoti hai, regulator nahi hota, aur gas expansion se propellant bahar nikalta hai.

Q11. Tank pressurization ke gas sources batao.

Ans: Ambient-temperature inert gas (He, N2, air), heated inert gas, chemical reaction se bani warm gas, cryogenic propellant ka evaporation, hypergolic direct injection, aur cryogenic ki self-pressurization.

Q12. Chemical equilibrium kya hota hai?

Ans: Reversible reaction me forward aur reverse reaction ki rate barabar ho jaaye to concentrations badalna band ho jaate hain. Combustion chamber me ye lagbhag constant pressure par hota hai aur gases Dalton's law follow karti hain.

Dhyan do

Unit 4 ke syllabus me turbo pump feed systems aur engine cycles, injectors, thrust chamber aur uska cooling, cryogenic propulsion system, hybrid rockets bhi likhe hain, par repo ke Unit 4 ke 2 PDFs (L1 aur L2) me ye topics abhi nahi aaye hain. Isliye is PDF me sirf wahi hai jo repo ke slides me tha.